

Evaluating the Effectiveness of User Awareness and Front-End Security Measures in Online Banking

Title of Research Project: Evaluating the Effectiveness of User Awareness and Front-End Security Measures in Online Banking

Principal Investigator: Shavindri Kuweju

Dear Participant,

I am a postgraduate student in the Cardiff School of Technologies at Cardiff Metropolitan University conducting research as part of my MSc Computing and Information Technology dissertation.

The aim of this study is to assess the cybersecurity awareness level of online banking users and their attitudes towards the cybersecurity protection features from the front-end of the online banking service, including Multi-Factor Authentication (MFA), One-Time Passwords (OTPs), biometric authentication, and security alerts, in the context of new cyber threats such as phishing, social engineering, and deep fake scams, as well as threats based on Artificial Intelligence. The results will be used for recommendations to raise consumers' awareness and enhance front-end cybersecurity in online banking.

There is no obligation to participate in this study. It will take about 8-10 minutes to complete the questionnaire. You can stop the survey at any time without giving it another thought by just closing the survey.

Your responses will remain anonymous and confidential. No personally identifiable information will be collected, and all data will be used solely for academic research purposes in accordance with Cardiff Metropolitan University's research ethics guidelines and the UK General Data Protection Regulation (UK GDPR).

By submitting this questionnaire, you confirm that you have read and understood the participant information provided, are aged 18 years or above, and voluntarily consent to participate in this research.

Thank you for your valuable time and participation.

Kind regards,
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* Indicates required question

1. Please confirm the following before proceeding with the survey: *

Please tick all of the above boxes to continue to the questionnaire.

Tick all that apply.

- ☐ I have read and understood the information provided about this research study.
- ☐ I understand that my participation is entirely voluntary, and I am free to stop completing the survey at any time before submitting my responses.
- ☐ I voluntarily agree to participate in this research study and provide my informed consent.

Untitled section

2. What is your age? *

Mark only one oval.

- ☐ 18-24
- ☐ 25-34
- ☐ 35-44
- ☐ 45-54
- ☐ 55+

3. Gender *

Mark only one oval.

- ☐ Male
- ☐ Female
- ☐ Prefer not to say

4. Highest Level of Education *

Mark only one oval.

☐ Secondary School

☐ Diploma

☐ Bachelor's Degree

☐ Master's Degree

☐ Doctorate

☐ Other:

Untitled section

5. How often do you use online banking? *

Mark only one oval.

☐ Daily

☐ Weekly

☐ Monthly

☐ Rarely

6. Which online banking platform do you use most frequently? *

Mark only one oval.

☐ Mobile Banking App

☐ Internet Banking Website

☐ Both equally

7. I can recognise phishing emails pretending to be from a bank *

Mark only one oval.

1 2 3 4 5

Stro ☐ ☐ ☐ ☐ ☐ Strongly Disagree

8. I know that legitimate banks never ask customers to share OTPs or passwords. *

Mark only one oval.

1 2 3 4 5

Stro ☐ ☐ ☐ ☐ ☐ Strongly Disagree

9. I understand how Multi-Factor Authentication (MFA) protects online banking accounts *

Mark only one oval.

1 2 3 4 5

Stro ☐ ☐ ☐ ☐ ☐ Strongly Disagree

10. I know how to verify that a banking website is genuine before entering my login credentials. *

Mark only one oval.

1 2 3 4 5

Stro ☐ ☐ ☐ ☐ ☐ Strongly Disagree

11. I understand the risks of using public Wi-Fi for online banking. *

Mark only one oval.

1 2 3 4 5

Stro ☐ ☐ ☐ ☐ ☐ Strongly Disagree

12. I am aware of AI-generated scams and deepfake fraud targeting banking customers. *

Mark only one oval.

1 2 3 4 5

Stro ☐ ☐ ☐ ☐ ☐ Strongly Disagree

13. I know what actions to take if I suspect my online banking account has been compromised. *

Mark only one oval.

1 2 3 4 5

Stro ☐ ☐ ☐ ☐ ☐ Strongly Disagree

Untitled section

14. Have you ever received a phishing email pretending to be from a bank? *

Mark only one oval.

☐ Yes

☐ No

☐ Not Sure

15. Have you ever received a suspicious SMS claiming to be from a bank? *

Mark only one oval.

- ☐ Yes
- ☐ No
- ☐ Not Sure

16. Have you ever received a phone call claiming to be from your bank asking for personal information? *

Mark only one oval.

- ☐ Yes
- ☐ No
- ☐ Not Sure

17. Have you ever clicked on a suspicious banking link? *

Mark only one oval.

- ☐ Yes
- ☐ No
- ☐ Not Sure

18. Have you ever experienced online banking fraud? *

Mark only one oval.

- ☐ Yes
- ☐ No

19. If yes, how did you identify or respond to the phishing attempt?
(Select all that apply)

Tick all that apply.

- ☐ I noticed suspicious links or website URLs.
- ☐ The email contained spelling or grammar mistakes.
- ☐ The sender's email address looked suspicious.
- ☐ The message requested sensitive information (passwords, PINs, OTPs).
- ☐ My bank warned me about similar scams.
- ☐ I ignored and deleted the email.
- ☐ I reported it to my bank.
- ☐ I clicked the link before realizing it was suspicious.
- ☐ I was unsure whether it was a phishing attempt.
- ☐ Other: _____

Untitled section

20. You receive an email claiming your bank account will be suspended unless you ^{*} click a link immediately.

What would you do?

Mark only one oval.

- ☐ Click the link immediately
- ☐ Ignore it completely
- ☐ Verify the request using the bank's official website or customer service
- ☐ Reply to the email

21. Someone claiming to be from your bank asks for your OTP to verify your identity. *

What would you do?

Mark only one oval.

- ☐ Share the OTP
- ☐ Ask why it is needed
- ☐ End the call and contact the bank through official channels
- ☐ Ignore the request but continue the conversation

22. You receive a security alert saying someone logged into your account from another device. *

What is your first action?

Mark only one oval.

- ☐ Ignore it
- ☐ Change your password immediately and contact the bank
- ☐ Wait to see if it happens again
- ☐ Delete the notification

23. You receive a voice call that sounds exactly like your bank manager asking you to urgently approve a transaction. *

What would you do?

Mark only one oval.

- ☐ Approve the transaction
- ☐ Verify the request using the bank's official contact details
- ☐ Share your banking details
- ☐ End the call without checking

Untitled section

24. Multi-Factor Authentication (MFA) improves online banking security. *

Mark only one oval.

	1	2	3	4	5	
Stro	☐	☐	☐	☐	☐	Strongly Disagree

25. Biometric authentication (Face ID/Fingerprint) provides better protection than passwords alone. *

Mark only one oval.

	1	2	3	4	5	
Stro	☐	☐	☐	☐	☐	Strongly Disagree

26. Bank security alerts help prevent cyberattacks. *

Mark only one oval.

	1	2	3	4	5	
Stro	☐	☐	☐	☐	☐	Strongly Disagree

27. Current front-end security measures provide adequate protection against emerging cyber threats. *

Mark only one oval.

	1	2	3	4	5	
Stro	☐	☐	☐	☐	☐	Strongly Disagree

28. AI-based fraud detection systems improve online banking security. *

Mark only one oval.

	1	2	3	4	5	
Stro	☐	☐	☐	☐	☐	Strongly Disagree

29. I use a unique password for my online banking account. *

Mark only one oval.

- ☐ Never
- ☐ Rarely
- ☐ Sometimes
- ☐ Always

30. I enable Multi-Factor Authentication whenever it is available. *

Mark only one oval.

- ☐ Never
- ☐ Rarely
- ☐ Sometimes
- ☐ Always

31. I regularly update my banking application. *

Mark only one oval.

- ☐ Never
- ☐ Rarely
- ☐ Sometimes
- ☐ Always

32. I verify website authenticity before entering my banking credentials. *

Mark only one oval.

- ☐ Never
- ☐ Rarely
- ☐ Sometimes
- ☐ Always

Untitled section

33. I understand how biometric authentication (e.g., Face ID or fingerprint recognition) works when I log in to my online banking account. *

Mark only one oval.

	1	2	3	4	5	
Stro	☐	☐	☐	☐	☐	Strongly Disagree

34. I understand how my biometric data (e.g., Face ID or fingerprint) is stored and protected when I use biometric authentication for online banking. *

Mark only one oval.

1 2 3 4 5

Stro ☐ ☐ ☐ ☐ ☐ Strongly Disagree

35. I am concerned about the privacy risks associated with using biometric authentication for online banking. *

Mark only one oval.

1 2 3 4 5

Stro ☐ ☐ ☐ ☐ ☐ Strongly Disagree

36. I trust that my biometric data is securely protected when I use biometric authentication for online banking. *

Mark only one oval.

1 2 3 4 5

Stro ☐ ☐ ☐ ☐ ☐ Strongly Disagree

37. Banks should provide clearer information about how biometric data is used and protected when customers use biometric authentication for online banking. *

Mark only one oval.

1 2 3 4 5

Stro ☐ ☐ ☐ ☐ ☐ Strongly Disagree

38. Concerns about biometric privacy would affect my willingness to use biometric authentication for online banking. *

Mark only one oval.

1 2 3 4 5

Strongly Disagree

39. Which biometric authentication method do you currently use for online banking? *

Tick all that apply.

- ☐ Face ID / Facial Recognition
☐ Fingerprint Recognition
☐ Voice Recognition
☐ I do not use biometric authentication
☐ Other: _____

40. In your opinion, what is the biggest concern when using biometric authentication for online banking? *

Tick all that apply.

- ☐ Privacy of biometric data
☐ Risk of biometric data being stolen
☐ Lack of understanding of how biometric data is used
☐ Accuracy of biometric authentication
☐ I have no concerns
☐ Other: _____

Thank You for Your Participation

Thank you for taking the time to complete this questionnaire.

Your participation is greatly appreciated and will make a valuable contribution to this research on cybersecurity awareness and front-end security measures in online banking.

Your responses have been submitted anonymously and will be treated confidentially. The information you have provided will be used solely for academic research purposes as part of an MSc Computing and Information Technology dissertation at Cardiff Metropolitan University.

Shavindri Kuweju

MSc Computing and Information Technology
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Cardiff Metropolitan University

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